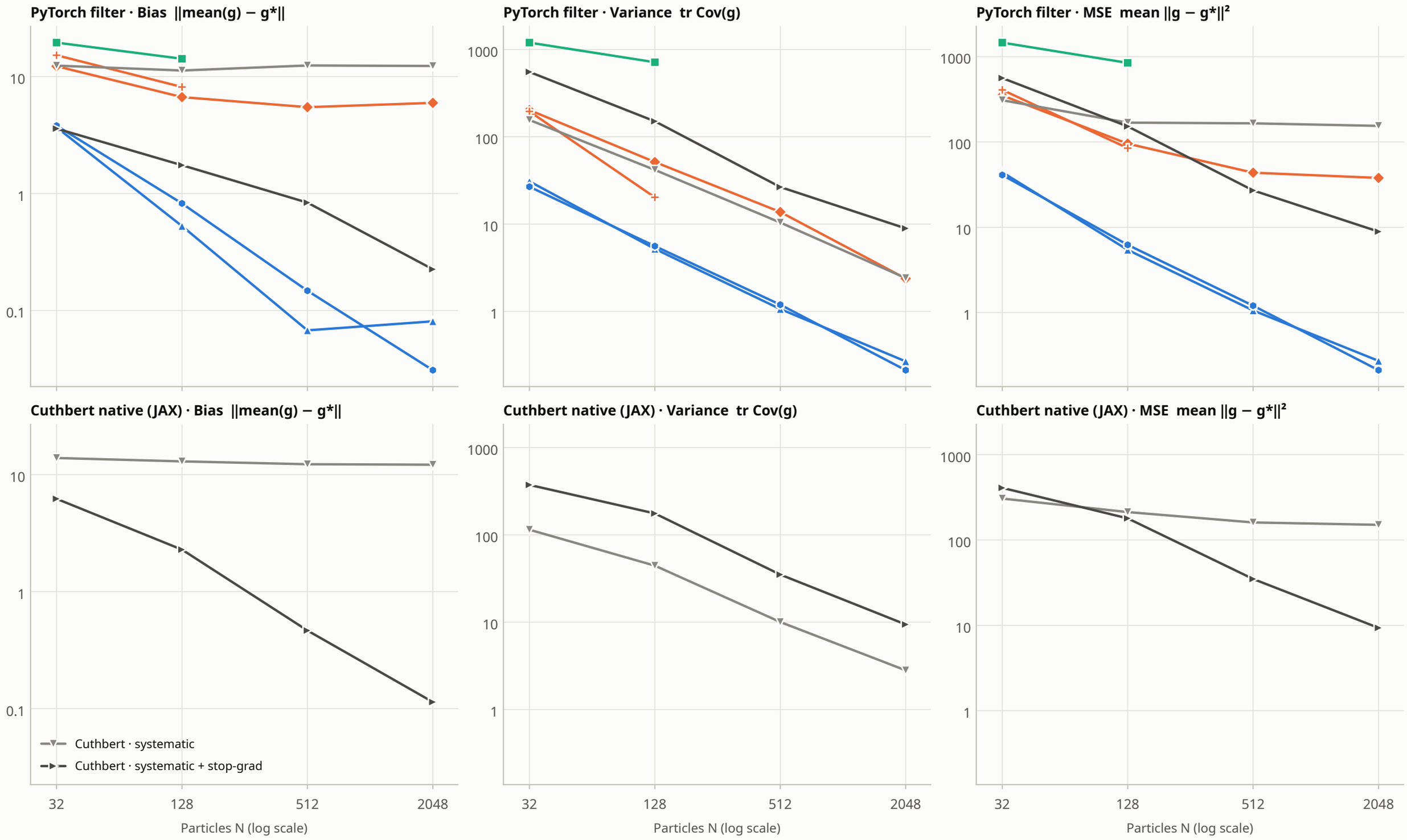
Particle-filter log-likelihood gradients against the exact Kalman gradient

Cuthbert tutorial random-walk model and saved dataset (T = 50) · 100 seeds per N (Gumbel and PyDPF OT: 10, N ≤ 128) · log scales

Local · eigen
PyDPF · soft
PyDPF · OT
PyDPF · systematic + stop-grad
Local · Gaussian
Local · Gumbel-softmax
PyDPF · systematic



Source: torch_native_summary.csv, cuthbert_native_summary.csv. Gradient w.r.t. (log q, log r). PyDPF soft and torchfilter soft give identical values; only PyDPF is drawn.

Resample threshold 0.7N, as in the Cuthbert tutorial. No confidence intervals are computed for these summaries.